

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EE/ME/CL/CE/IT/EC)

December 2013

PY 101 Engineering Physics

Date: 04.12.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Choose the correct answer for the following questions.

[07]

- (a) Choose the correct statement.
- (i) Sound travels faster in vacuum (ii) Sound waves are transverse wave
- (iii) Sound travels faster solid than liquid vacuum (iv) all are correct
- (b) If the error in the measurement of radius of a sphere is 1% then what is the error in the measurement of volume?
- (i) 1 % (ii) 3 %
- (iii) 2 % (iv) 4 %
- (c) What are the qualities by which a musical sound is characterized?
- (i) Pitch, loudness and quality (ii) Pitch, loudness and amplitude
- (iii) loudness, audibility and frequency (iv) loudness, quality and intensity
- (d) Ultrasonic waves are those waves which _____.
- (i) human being cannot hear (ii) have high velocity
- (iii) have large amplitude (iv) none of the above
- (e) Optical fibre communication is based on the phenomenon of _____.
- (i) scattering (ii) diffraction
- (iii) refraction (iv) total internal reflection
- (f) He-Ne gas laser give _____ colored light.
- (i) pink (ii) red
- (iii) orange (iv) green
- (g) Laser light is considered to be coherent because it consists of _____.
- (i) many wavelengths (ii) uncoordinated wavelengths
- (iii) coordinated with same wavelengths (iv) divergent beams

- Q - 2 (a) "Data without uncertainty cannot be relevant scientifically" How do you justify this statement? [3]
- (b) Define intensity level. The intensity of sound produced by a machine is 0.2 W/m^2 . Calculate its intensity level in decibel. [4]
- (c) Explain how ultrasound is produced by magnetostriction method. Discuss three applications of ultrasonic waves. [7]

OR

- Q - 2 (a) A student measures two quantities a & b and obtained the results $a = 11.5 \pm 0.2$ and $b = 25.4 \pm 0.2$. Find the product $q = ab$ giving both absolute and percentage uncertainties. [3]
- (b) Why are ultrasonic waves not audible to humans? Mentions the properties of ultrasonic waves. [4]
- (c) What are the different sources of noises in a building? Define echelon effect, echo, reverberation time and resonance. [7]
- Q - 3 (a) An ultrasonic beam is used to determine the length of a swimming pool. If the frequency of the fundamental note is 50 Hz , what is the length of the pool if the velocity of ultrasonic wave in water is 1330 m/s ? [3]
- (b) Establish the relation between Einstein's coefficients A and B. [4]
- (c) State and explain the principle of holography. Why is a laser beam needed for it? Give two applications of holography. [7]

OR

- Q - 3 (a) Calculate the numerical aperture and the acceptance angle of optical fiber. Given that refractive index of core = 1.62 and of cladding = 1.52. [3]
- (b) Define : (i) Stimulated emission, (ii) Stimulated absorption, (iii) Population Inversion and (iv) Metastable state [4]
- (c) What are the types of optical fibers? Outline some merits of optical fiber communications. [7]

SECTION - II

- Q - 4 Choose the correct answer for the following questions. [07]
- (a) A normal metal is converted into a superconductor below _____.
 (i) critical magnetic field (ii) critical temperature
 (iii) critical current density (iv) all the above
- (b) Superconductors are perfect _____ materials.
 (i) diamagnetic (ii) paramagnetic
 (iii) ferromagnetic (iv) dielectric
- (c) What is the longest wavelength that can be analyzed by a rock salt crystal of spacing $d = 2.82 \text{ \AA}$ in the second order?
 (i) $2.82 \text{ \AA}$ (ii) $5.64 \text{ \AA}$
 (iii) $1.41 \text{ \AA}$ (iv) $3.82 \text{ \AA}$
- (d) _____ has (have) wavelengths that are shorter than X-rays.
 (i) Gamma-rays (ii) Ultraviolet
 (iii) Infrared radiation (iv) both (ii) and (iii)

- (e) The number of Bravais lattices is _____.
 (i) 7 (ii) 14
 (iii) 10 (iv) 32
- (f) What is the frequency of the electromagnetic waves radiated from a Josephson junction, if the voltage drop at the junction is $650\mu\text{V}$?
 (i) 314GHz (ii) 320GHz
 (iii) 414GHz (iv) 341GHz
- (g) If the number of lattice points per unit cell is one, then it is called ____ unit cell.
 (i) primitive (ii) cubic
 (iii) non-primitive (iv) none of the above

- Q - 5 (a) An X-ray tube operated at 30 kV emits a continuous X-ray spectrum with a short wavelength limit $\lambda_{\text{min}} = 0.414\text{\AA}$. Calculate Planck's constant h . [3]
 [Given, $c = 3 \times 10^8\text{ m/s}$; $e = 1.6 \times 10^{-19}\text{ C}$].
- (b) Explain the principle of X-ray diffraction. How is it used to determine the unit cell dimensions? [4]
- (c) What is superconductivity? Calculate the critical magnetic field at 1°K for mercury if its transition temperature is 4.15°K and critical magnetic field at 0°K is $H_c = 3.27 \times 10^6\text{ A/m}$. [7]

OR

- Q - 5 (a) What is the significance of critical temperature, critical magnetic field, and critical current density for superconductors? [3]
- (b) Explain: (i) Hard superconductor, (ii) Soft superconductor, (iii) isotope effect and (iv) Josephson ring [4]
- (c) What is Hall effect? The electrical resistivity of a metal is $4 \times 10^{-8}\text{ }\Omega\text{m}$. The cross-sectional dimension of the specimen is $25\text{ mm} \times 25\text{ mm}$. A current of 30 A is allowed to flow at right angles to the magnetic field of 0.75 T and a Hall voltage of $-1.26 \times 10^{-7}\text{ V}$ is observed across the ends of the specimen. Identify the type of material and calculate the electron charge concentration and the mobility in the material. [7]
- Q - 6 (a) How are the solid materials classified from the crystal structure point of view? [3]
- (b) Define interplanar distance. The lattice constant of a cubic crystal is $2.5\text{\AA}$. Find the lattice spacing for the following planes in the lattice: $(1\ 0\ 1)$ and $(3\ 1\ 0)$. [4]
- (c) What is nanotechnology? Explain briefly how nanotechnology will change the world? [7]

OR

- Q - 6 (a) Sketch the planes for the Miller indices of (100) , (110) and (111) [3]
- (b) Define crystal lattice, unit cell, atomic packing factor and coordination number. [4]
- (c) What is nanoscale? Discuss briefly the reasons for the difference in material properties by applying nanoscale. [7]